Human Activity Recognition (HAR) Project |Phases 1 & 2 Report

Course: Supervised Learning

Project ID: 6

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# 1. Project Objective

The goal of this project is to develop a robust supervised learning pipeline capable of classifying human physical activities (Walking, Walking_Upstairs, Walking_Downstairs, Sitting, Standing, Laying) using tri-axial signal data from smartphone accelerometers and gyroscopes. This report covers the foundational data engineering (Phase 1) and the analytical modeling (Phase 2).

# 2. Phase 1: Data Engineering & Foundation

> **Focus:** Data Acquisition, Cleaning, and Structural Preparation.

In the initial phase, the UCI HAR dataset was downloaded and programmatically extracted. Key technical steps included:

- **Data Loading:** Mapping the 561-feature vector from raw text files (`X_train.txt`) and aligning them with their respective activity labels (`y_train.txt`).
- **Metadata Mapping:** Utilizing `features.txt` to assign descriptive column names to the dataset, ensuring the data is interpretable for later analysis.
- **Serialization:** To optimize Phase 2, the data was cleaned of potential null values and exported to `train_features.csv` and `test_features.csv`, establishing a stable starting point for modeling.

# 3. Phase 2: Intelligent Data Analysis (IDA) & Modeling

> **Focus:** Exploratory Analysis and Supervised Model Benchmarking.

## 3.1 Intelligent Data Analysis (IDA)

Using `ida.py`, we performed a deep dive into the sensor signals:

- **Signal Patterns:** Visualized the first three features (X, Y, Z axes), identifying clear signature differences between dynamic movements and static positions.
- **Feature Correlation:** Generated a heatmap for the first 20 features, highlighting strong dependencies between sensor axes, which informs future dimensionality reduction needs.
- **Statistical Distribution:** Confirmed normalized distributions for several features through histogram and KDE plots, satisfying the assumptions for probabilistic models.

### 3.2 Supervised Learning Implementation

The core logic in `ml_modes.py` evaluates several classification paradigms:

| Model Type | Implementation | Role in Project |
|---|---|---|
| **Instance-Based** | K-Nearest Neighbors (KNN) | Baseline for distance-based similarity in high dimensions. |
| **Probabilistic** | Gaussian Naive Bayes | Tests the efficiency of Gaussian distribution assumptions. |
| **Ensemble (Bagging)** | Random Forest | Handles non-linear relationships and reduces overfitting. |
| **Ensemble (Boosting)** | Gradient Boosting | Optimized for high accuracy through sequential error reduction. |

```
Directory/ ├── project.ipynb # Integrated Research (Phases 1 & 2 combined) ├── ida.py #
Exploratory Visualizations & Statistical Checks ├── ml_modes.py # Machine Learning
Fipelinraining/Evaluation) ├── train_features.csv # Structured Training
Data └── test_features.csv # Structured Testing Data
```

## 4. Current Conclusions & Next Steps

Phase 1 and 2 have successfully converted raw sensor signals into a predictive engine. While classical models like Random Forest show high performance, Phase 3 will explore **Deep Learning** (CNN/LSTM) to capitalize on the raw inertial signals, potentially capturing temporal patterns that classical feature-based models may overlook.